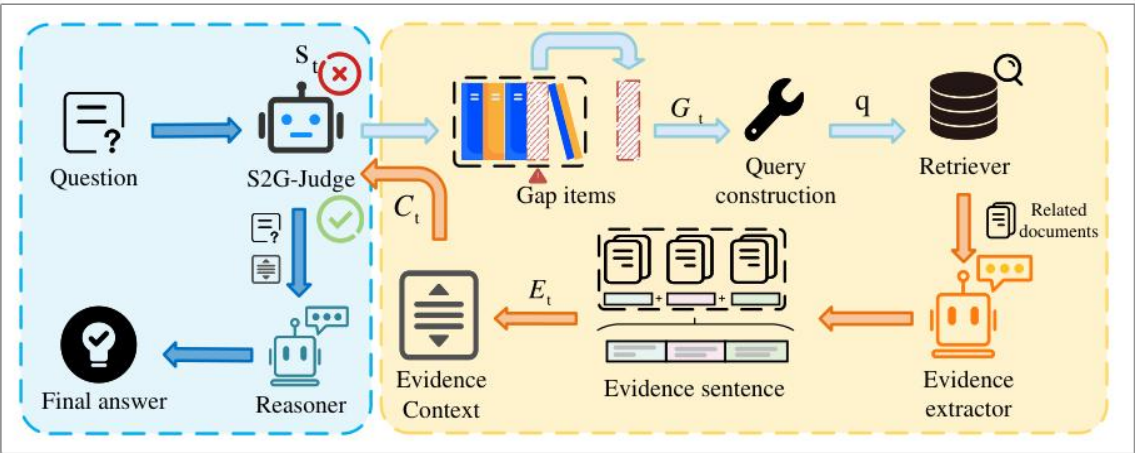


S2G-RAG: Structured Sufficiency and Gap Judging for Iterative Retrieval-Augmented QA

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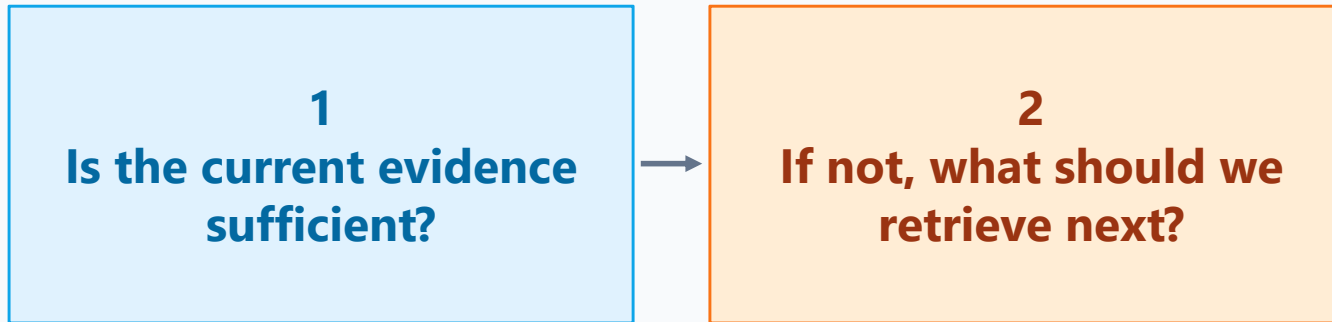
Presenter: Minghan Li



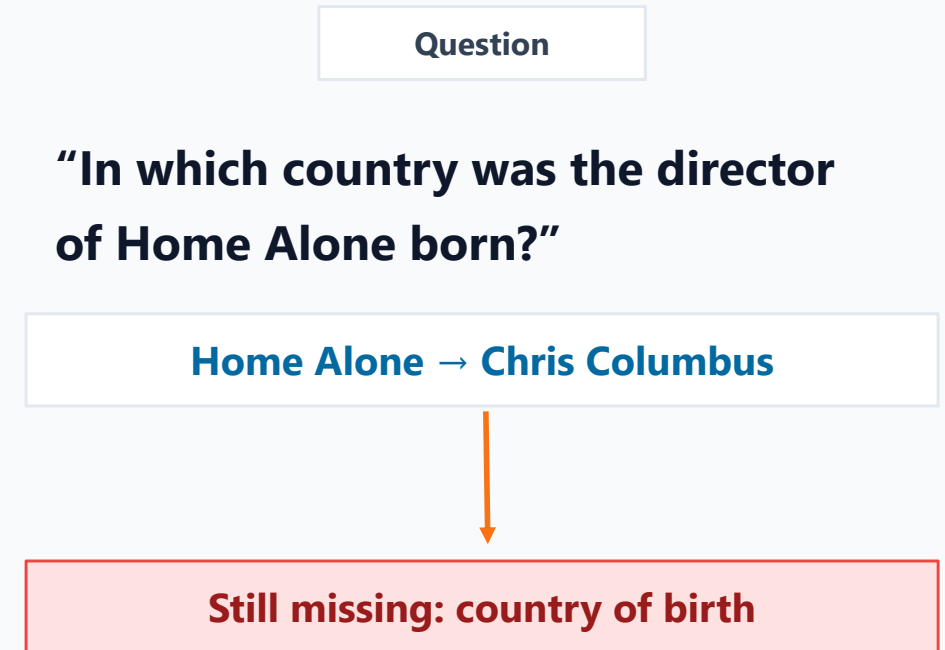
Make iterative RAG control explicit and auditable

Iterative RAG has a control bottleneck

At every retrieval turn, the system must decide:



Multi-hop QA often requires a chain of evidence, not just one retrieved document.



Two common failure modes

Stops too early

Answers from incomplete evidence chains.

Risk: high-confidence but unsupported answers.

Continues blindly

Retrieves without a clear target.

Risk: redundant or distractor-heavy context.

What we need

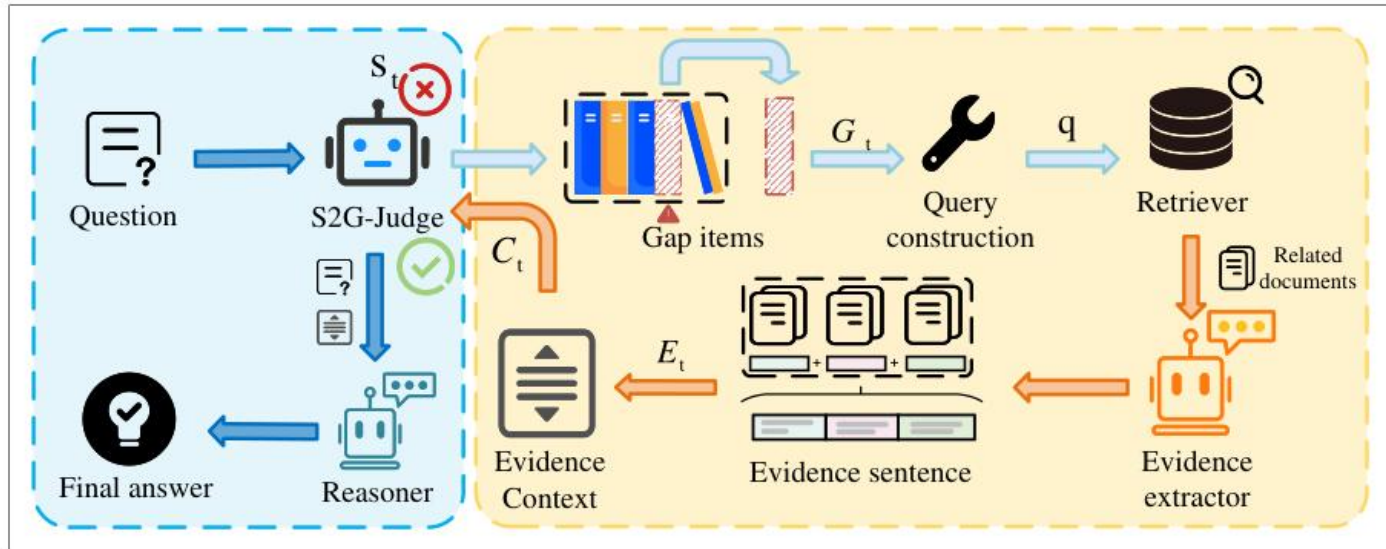
An explicit turn-level state:

- whether evidence is adequate
- what concrete gap remains

Control should be structured, diagnosable, and separated from answer generation.

Core idea

S2G-RAG: explicit sufficiency + gap control



S2G-Judge reads:

question q
+ evidence context C_t

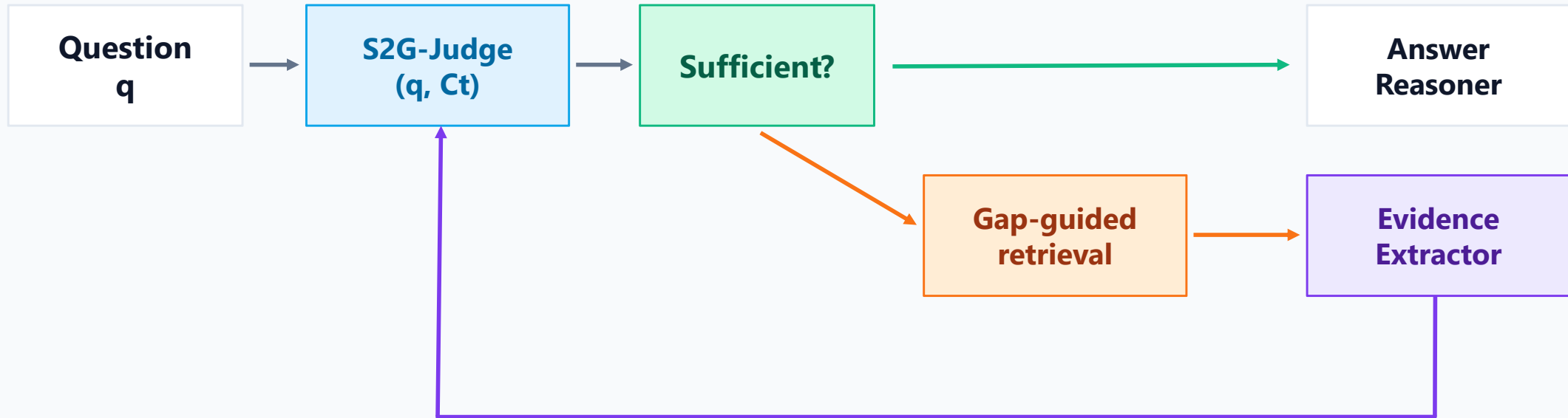
and predicts:

sufficiency
true / false

structured gaps
if false

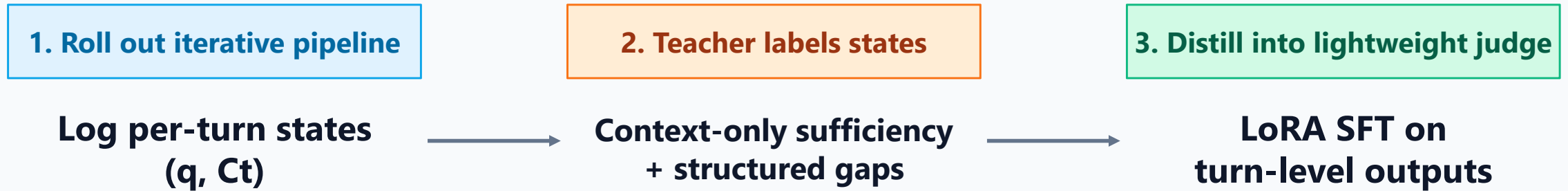
The answer reasoner is decoupled
from retrieval control.

Judge first, retrieve only when needed



If sufficient: answer.
If not: retrieve with gaps, extract evidence sentences, update C_t .

Training S2G-Judge from realistic execution traces



Why traces?

They expose the judge to intermediate contexts produced by real multi-turn retrieval, including redundancy and distractors.

Context-only labeling prevents sufficiency judgments from relying on parametric knowledge.

Structured gap items specify what is missing

Gap item schema

category

bridge_entity / attribute / relation /
evidence_span / other

target

entity being queried

slot

missing attribute or relation

description

short clarification

Worked example

Current evidence:

Home Alone → director → Chris Columbus

Gap item:

category = attribute
target = Chris Columbus
slot = country of birth

Constructed query:

“Chris Columbus country of birth”

Not just “retrieve more” — retrieve the missing fact.

Do not accumulate full documents

Iterative retrieval naturally grows long and noisy unless evidence memory is controlled.

Raw document concatenation

- long context
- redundant passages
- distractor-heavy retrieval
- unstable judging and reasoning



S2G-RAG Evidence Context

- selected salient sentences
- gap-aware extraction
- provenance-preserving
- compact memory across turns

Gap-aware extraction controls context growth across retrieval turns.

Does structured sufficiency and gap judging work?

Evaluation questions

Does it improve QA accuracy?

Which component matters most?

Does it reduce noisy context?

Experimental setup

Tasks

TriviaQA
single-hop

HotpotQA
multi-hop

2WikiMultiHopQA
multi-hop

Retrieval settings

Sparse retrieval
BM25 / Pyserini

Dense retrieval
E5-base-v2

$T \leq 4$ turns
top-k = 6

Models

Llama-3-8B

answer reasoner

Llama-3.2-3B

S2G-Judge backbone

GPT-4o-mini

teacher labels

S2G-RAG improves QA under sparse and dense retrieval

Method	Reasoner / Retriever	TriviaQA		HotpotQA		2Wiki	
		EM	F1	EM	F1	EM	F1
<i>Panel A: Sparse retrieval (BM25)</i>							
NaiveGen	Llama3-8B / –	55.7	63.1	20.6	28.4	26.4	33.9
IR-CoT	Llama3-8B / BM25	56.9	68.9	28.6	41.5	23.5	32.4
SIM-RAG	Llama3-8B / BM25	70.7	75.6	32.7	43.3	34.1	40.2
S2G-RAG (ours)	Llama3-8B / BM25	72.0	77.9	43.3	56.5	41.7	48.6
<i>Panel B: Dense retrieval (E5)</i>							
Standard RAG	Llama3-8B / E5	58.8	68.3	25.1	35.3	10.6	21.0
Self-RAG	Llama3-8B / E5	38.2	53.4	17.1	29.6	12.1	25.1
FLARE	Llama3-8B / E5	55.8	63.2	19.7	28.0	25.8	33.9
ReSP*	Llama3-8B / E5	–	–	–	47.2	–	38.3
RAG-Critic	Llama3-8B / E5	65.0	75.9	40.0	51.2	27.9	34.0
S2G-RAG (ours)	Llama3-8B / E5	71.1	78.0	42.0	53.5	39.0	45.3

Largest gains appear on multi-hop datasets.

+10.6 EM

HotpotQA vs. SIM-RAG (BM25)

+13.2 F1

HotpotQA vs. SIM-RAG (BM25)

+7.6 EM

2Wiki vs. SIM-RAG (BM25)

+8.4 F1

2Wiki vs. SIM-RAG (BM25)

What matters most? S2G-Judge

Ablation on HotpotQA under BM25

Variant	EM	F1
Full S2G-RAG	43.3	56.5
w/o SFT (Untrained Judge)	39.2 (−4.1)	50.8 (−5.7)
w/o S2G-Judge	27.5 (−15.8)	37.6 (−19.0)
w/o Extractor	39.5 (−3.8)	52.5 (−4.0)

Removing S2G-Judge causes the largest drop
−15.8 EM / −19.0 F1 on HotpotQA

Explicit retrieval control is the core driver of the improvement.

without judge

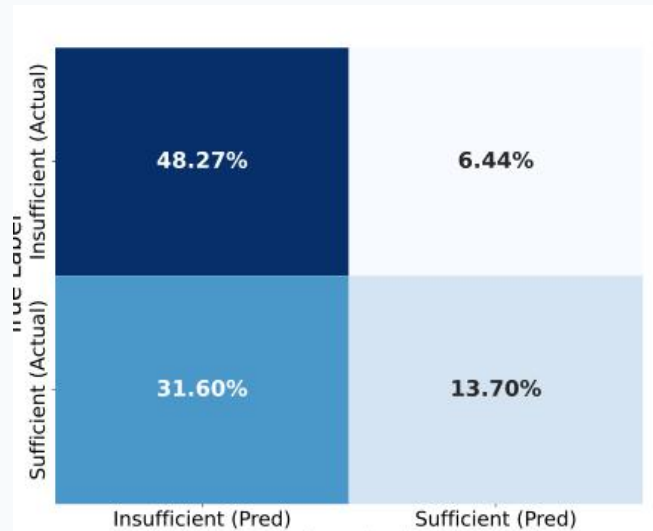


no sufficiency check
no gap signal

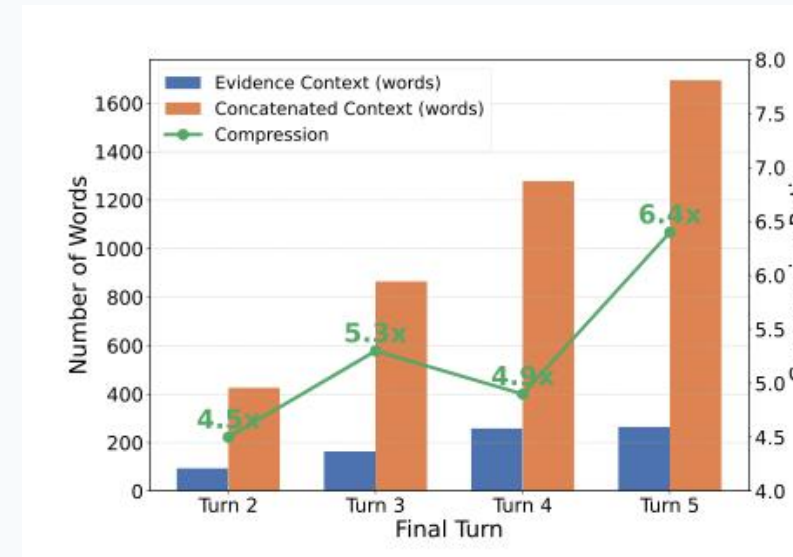
The trained judge decides both when to stop and what to retrieve next.

Less noise, better control

S2G-RAG is conservative about stopping and compact about evidence memory.



False-positive rate: 6.44%
Judge rarely over-claims sufficiency



Context compression: 4.5x–6.4x
Less redundancy across turns

Takeaways

1

Make iterative RAG control explicit

2

Use structured gaps to guide next-hop retrieval

3

Keep evidence compact and auditable

Thank you!